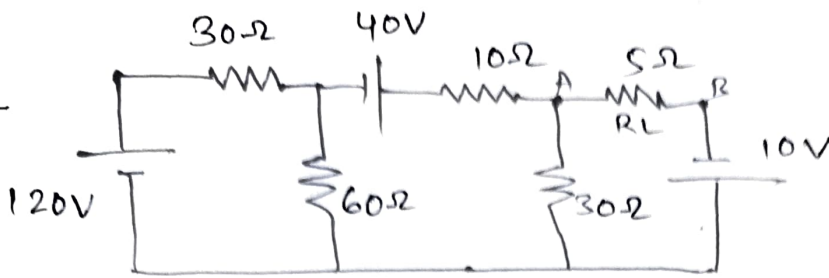
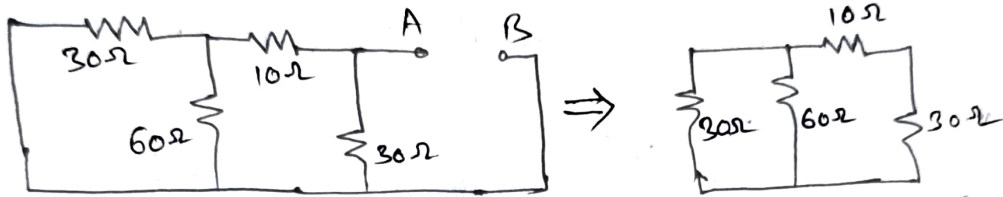


Q.1



Step 1 :- Calculation of R_N

Remove R_L & short ckt all volg sources
Resultant ckt



$$30\Omega \parallel 60\Omega = \frac{30 \times 60}{30 + 60} = 20\Omega$$

$$20\Omega \text{ in series with } 10\Omega = 20 + 10\Omega = 30\Omega$$

$$30\Omega \parallel 30\Omega = 15\Omega$$

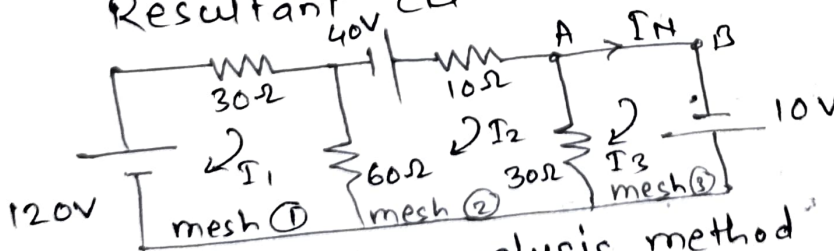
$$\therefore \boxed{R_N = 15\Omega}$$

(3m)

Step 2 : Calculation of I_N

Replacing R_L by a short ckt

Resultant ckt



Using mesh analysis method I_N is calculated:

Applying KVL to mesh 1

$$120 - 30I_1 - 60(I_1 - I_2) = 0$$

or

$$90I_1 - 60I_2 = 120$$

$$3I_1 - 2I_2 = 4 \quad \text{--- (1)}$$

Applying KVL to mesh 2

$$-60(I_2 - I_1) + 40 - 10I_2 - 30(I_2 - I_3) = 0$$

or

$$-60I_1 + 100I_2 - 30I_3 = 40$$

$$-2I_1 + 3.33I_2 - 1I_3 = 1.33 \quad \text{--- (2)}$$

(5m)

Applying KVL to mesh 3

$$-30(I_3 - I_2) + 10 = 0$$

$$\text{or } 30I_2 - 30I_3 = -10$$

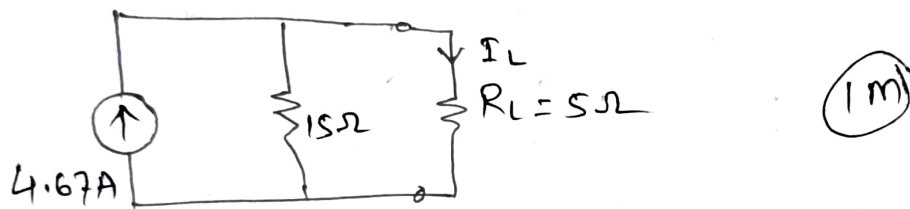
$$I_2 - I_3 = -0.33 \quad \text{--- (3)}$$

Solving eq's (1), (2) and (3)

$$I_3 = 4.67 \text{ Amp.}$$

$$I_H = I_3 \quad \therefore \boxed{I_H = 4.67 \text{ Amp}}$$

Step 3 :- Norton's equivalent circuit



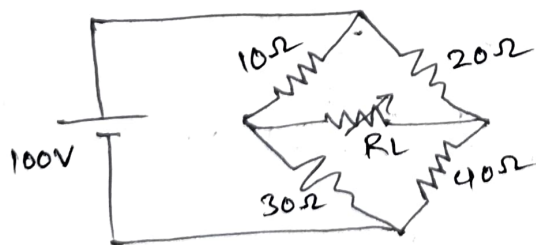
Step 4 :- Calculation of I_L

$$I_L = 4.67 \times \frac{15}{15+5}$$

$$\boxed{I_L = 3.50 \text{ Amp}}$$

OR

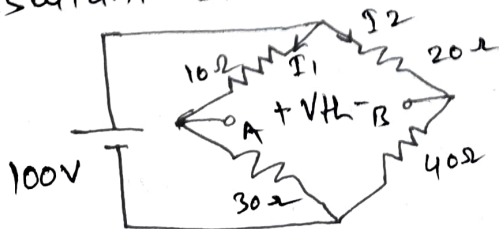
Q. 2



Step 1 :- Calculation of V_{th}

Remove R_L

Resultant ckt



(3)

$$I_1 = \frac{100}{10+30} = 2.5 \text{ A}$$

$$I_2 = \frac{100}{20+40} = 1.66 \text{ A}$$

From the ckt,

$$V_{th} + 10I_1 - 20I_2 = 0$$

$$\therefore V_{th} = 20I_2 - 10I_1$$

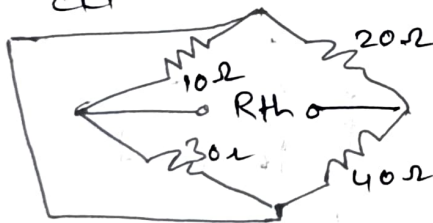
$$= 20(1.66) - 10(2.5)$$

$$V_{th} = 8.2 \text{ V}$$

Step 2 :- Calculation R_{th}

Open R_L & replace volg source by short ckt.

Resultant ckt



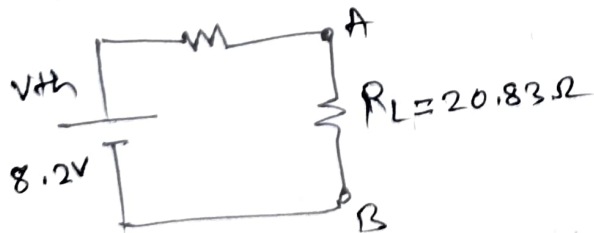
$$(10\Omega \parallel 30\Omega) + (20\Omega \parallel 40\Omega)$$

$$\therefore R_{th} = 20.83\Omega$$

Step 3 :- Calculation of R_L for Max. power transfer

For Max. power transfer $R_L = R_{th} = 20.83\Omega$

Step 4 :- Calculation of P_{max}

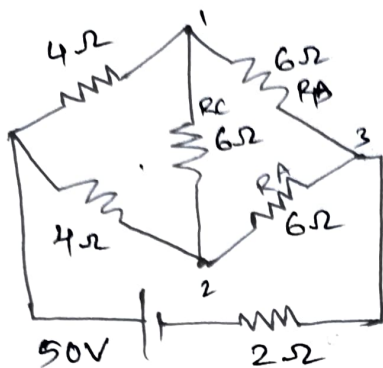


$$P_{max} = \frac{V_{th}^2}{4R_{th}}$$

$$= \frac{(8.2)^2}{4 \times 20.83}$$

$$P_{max} = 0.81 \text{ W}$$

Q. 2(a)



Delta network of $6\Omega, 6\Omega \text{ \& } 6\Omega$ can be converted into equivalent star n/w as:

using eqⁿ.

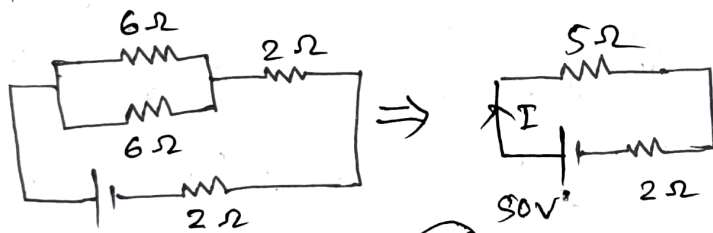
$$R_1 = \frac{R_B R_C}{R_A + R_B + R_C} = \frac{6 \times 6}{6 + 6 + 6} = \frac{36}{18}$$

$R_1 = 2\Omega$ — (1m)

Similarly for $R_2 \text{ \& } R_3$ we get

$R_2 = R_3 = 2\Omega$ — (1m)

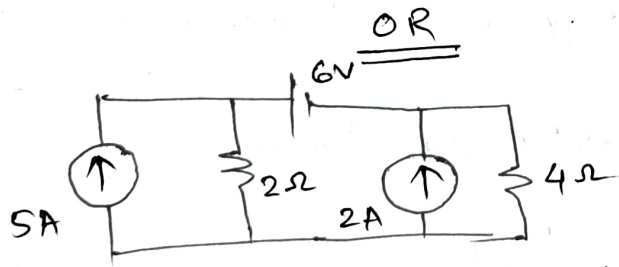
Simplifying n/w as



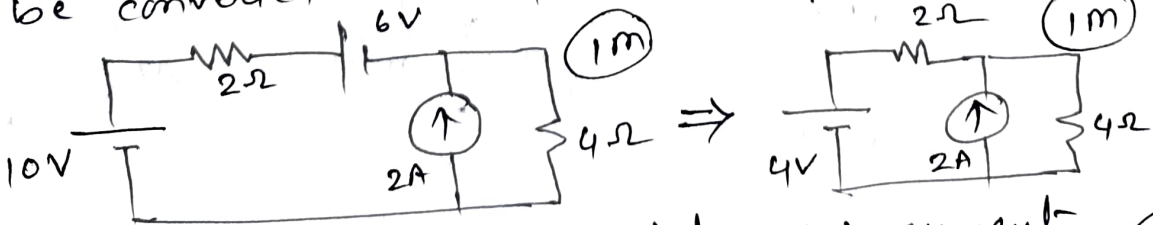
$\therefore I = \frac{50}{5+2}$ (1m)

$I = 7.14 \text{ Amp}$

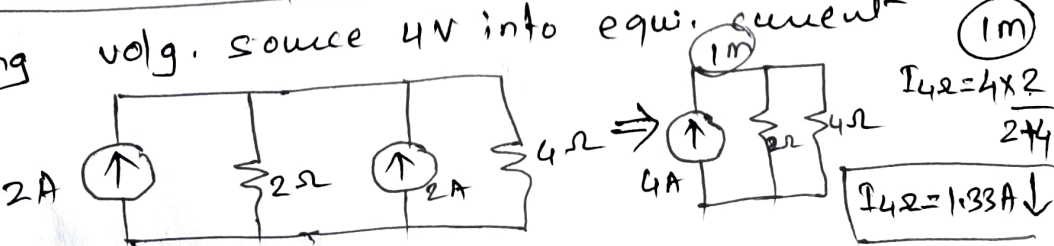
Q. 2(a)



Current source of 5A in || with 2Ω can be converted into equi. volg. source as:



Converting volg. source 4V into equi. current source

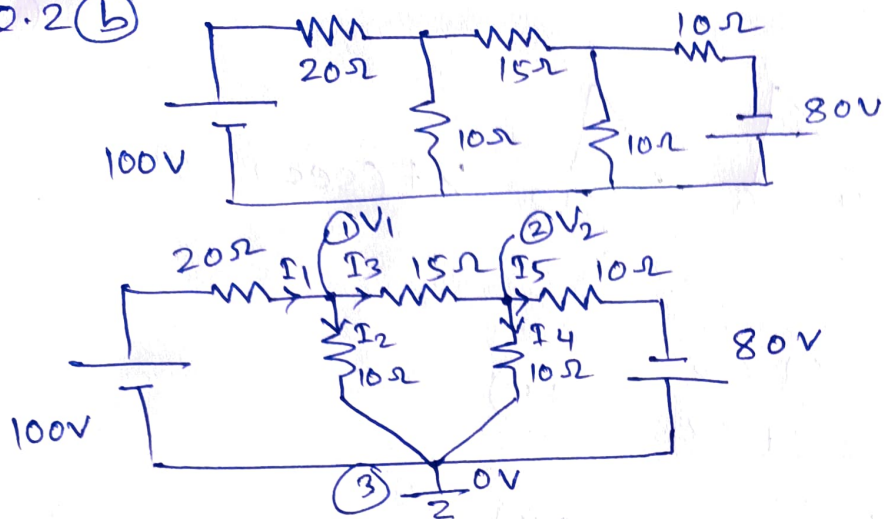


$I_{4\Omega} = 4 \times \frac{2}{2+4}$

$I_{4\Omega} = 1.33 \text{ A} \downarrow$

Q.2 (b)

(5)



Applying KCL at node 1

$$I_1 = I_2 + I_3$$

$$\frac{100 - V_1}{20} = \frac{V_1 - 0}{10} + \frac{V_1 - V_2}{15}$$

$$13V_1 - 4V_2 = 300 \quad \text{--- (1)}$$

Applying KCL at node 2

$$I_3 = I_4 + I_5$$

$$\frac{V_1 - V_2}{15} = \frac{V_2 - 0}{10} + \frac{V_2 - (-80)}{10} \quad \text{--- (2)}$$

$$V_1 - 4V_2 = 120$$

Solving eqs (1) & (2) we get

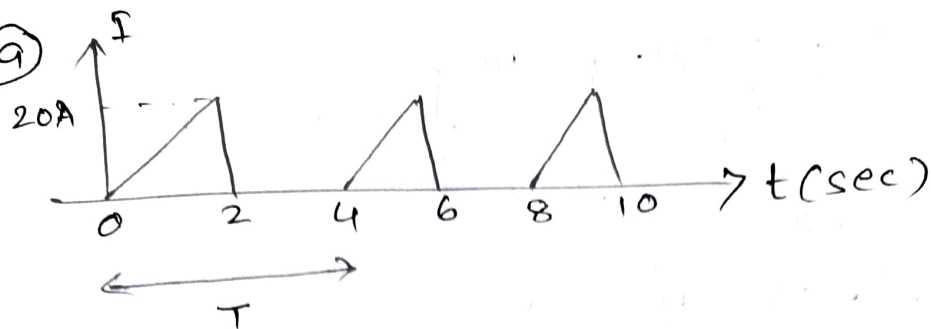
$$V_1 = 15V$$

$$V_2 = -26.25V$$

$$I_3 = I_{15\Omega} = \frac{V_1 - V_2}{15} = 2.75 \text{ Amp}$$

$$\therefore I_{15\Omega} = 2.75 \text{ Amp}$$

Q. 3a



$$i = 10t \quad 0 < t < 2 \quad (1m)$$

$$i = 0 \quad 2 < t < 4$$

$$I_{rms} = \sqrt{\frac{1}{4} \int_0^4 i^2 dt} \quad (1m)$$

$$I_{rms}^2 = \frac{1}{4} \int_0^4 i^2 dt \quad (1m)$$

$$= \frac{1}{4} \left[\int_0^2 i^2 dt + \int_2^4 i^2 dt \right] \quad (1m)$$

$$= \frac{1}{4} \int_0^2 (10t)^2 dt$$

$$= \frac{(10)^2}{4} \left[\frac{t^3}{3} \right]_0^2 \quad (1m)$$

$$\therefore I_{rms} = 8.16 \text{ Amp}$$

Q. 3b $L = 20 \text{ mH}$, $i = 10 \cos(1000t) \text{ Amp.}$
eq. of i can be written as:

$$i = 10 \sin(1000t + \frac{\pi}{2}) \quad (1m)$$

$$\omega = 1000 \text{ rad/sec} \quad (1m)$$

$$\phi = \frac{\pi}{2} \text{ or } 90^\circ$$

$$\text{rms value of current} = \frac{10}{\sqrt{2}} = 7.07 \text{ Amp} \quad (1m)$$

$$\text{Inductive reactance} = X_L = \omega L = 1000 \times 20 \times 10^{-3}$$

$$X_L = 20 \Omega \quad (1m)$$

$$V_L = I_{rms} X_L = (7.07) \times (20) \quad (1m)$$

$$V_L = 141.4 \text{ volts}$$